

1. What is the purpose of the “Applied Steps” pane in Power Query?
 - **Tracks your data transformations** Each step—like renaming columns, filtering rows, changing data types—is listed in order. This makes it easy to see how your data has evolved.
 - **Allows easy editing or removal** You can click on any step to review or modify it. If something breaks or needs adjustment, you don’t have to start over—just tweak the step.
 - **Supports debugging and troubleshooting** If your query throws an error, you can walk through the steps to pinpoint where things went wrong.
 - **Improves transparency and documentation** Anyone reviewing your query can understand exactly how the data was transformed, step by step.
 - **Enables dynamic updates** When the source data changes, Power BI re-applies these steps automatically, keeping your report up to date.
2. How do you remove duplicate rows in Power Query?

Open Power Query Editor

- In Power BI Desktop, go to **Home > Transform Data**
- #### **Select the Table**
- In the left pane, click on the query/table you want to clean
- #### **Choose Columns for Duplicate Check**
- If you want to remove duplicates based on all columns:
 - Go to the top ribbon and click **Remove Rows > Remove Duplicates**
 - If you want to remove duplicates based on specific columns:
 - Select those columns (hold **Ctrl** while clicking)
 - Then click **Remove Rows > Remove Duplicates**

Apply Changes

- Click **Close & Apply** to load the cleaned data into Power BI
3. What does the “Filter” icon do in Power Query?
 - **Displays filtering options** for the selected column
 - Allows you to:
 - Select specific values (e.g., only "Product A" and "Product B")
 - Apply number filters (e.g., greater than, less than, equals)
 - Apply date filters (e.g., before, after, between)
 - Apply text filters (e.g., contains, starts with, ends with)
 4. How would you rename a column from “CustID” to “CustomerID”?

Open Power Query Editor Go to **Home > Transform Data**

Locate the Column Find the column named **CustID** in your table

Rename the Column

- a. Right-click on the column header
- b. Select **Rename**
- c. Type **CustomerID** and press **Enter**

Alternatively:

- d. Select the column
- e. Go to the **Transform** tab in the ribbon
- f. Click **Rename**

5. What happens if you click "Close&Apply" in Power Query?

- **Closes Power Query Editor** It exits the data transformation interface and returns you to the main Power BI report view.
 - **Applies All Data Transformations** Every step you've made—filters, column renames, type changes, merges, etc.—is executed and saved.
6. Remove all rows where Quantity is less than 2.

Open Power Query Editor

- a. In Power BI Desktop, go to **Home > Transform Data**

Find the Quantity Column

- b. Locate the column named **Quantity** in your table

Apply the Filter

- c. Click the dropdown arrow next to the **Quantity** column header
- d. Choose **Number Filters > Greater Than or Equal To**
- e. Enter 2 and click **OK**

Apply Changes

- f. Click **Close & Apply** to load the filtered data into Power BI

7. Split the OrderDate column into separate "Year", "Month", and "Day" columns.

Open Power Query Editor

- In Power BI Desktop, go to **Home > Transform Data**

Ensure OrderDate is in Date Format

- If it's not already a date, right-click the column
- Choose **Change Type > Date**

Extract Year, Month, and Day

- Select the **OrderDate** column
- Go to the **Add Column** tab in the ribbon
- Use the following options:
 - **Date > Year > Year** → adds a column with the year
 - **Date > Month > Month** → adds a column with the month number
 - **Date > Day > Day** → adds a column with the day number

Rename the New Columns (Optional)

- Right-click each new column and rename them to:
 - Year
 - Month
 - Day

Apply Changes

- Click **Close & Apply** to load the updated data into Power BI

8. Replace all “Mouse” entries in the Product column with “Computer Mouse”

Open Power Query Editor

- a. Go to **Home > Transform Data**

Select the Product Column

- b. Click on the **Product** column to highlight it

Use the Replace Values Feature

- c. Go to the **Transform** tab in the ribbon

- d. Click **Replace Values**

- e. In the dialog box:

- i. **Value to Find:** Mouse

- ii. **Replace With:** Computer Mouse

- f. Click **OK**

Apply Changes

- g. Click **Close & Apply** to load the updated data into Power BI

9. Sort the table by OrderDate (newest first).

Open Power Query Editor

- Go to **Home > Transform Data**

Locate the OrderDate Column

- Find the column named **OrderDate**

Sort the Column

- Click the small dropdown arrow next to the column header
- Choose **Sort Descending** (this will place the newest dates at the top)

Apply Changes

- Click **Close & Apply** to load the sorted data into Power BI

10. How would you handle null values in the Price column?

1. Remove Rows with Null Price
2. Replace Nulls with a Default Value
3. Use Conditional Logic

11. Write custom M-code to add a column calculating TotalSpent = Quantity * Price.

Open Power Query Editor

- a. Go to **Home > Transform Data**

Add a Custom Column

- b. In the ribbon, select **Add Column > Custom Column**

- c. In the dialog box:

- i. **New column name:** TotalSpent

- ii. **Formula:**[Quantity] * [Price]

- iii. Click **OK**

Check the Data Type

- d. Make sure the new column is set to **Decimal Number** or **Whole Number** depending on your needs

- e. You can change it by right-clicking the column header and selecting **Change Type**

Apply Changes

- f. Click **Close & Apply** to load the updated data into Power BI

12. Group the table by Cust/D to show total spending per customer.

- **Open Power Query Editor**
- Go to **Home > Transform Data**
- **Make Sure You Have a TotalSpent Column**
- If not already created, add it:
 - Go to **Add Column > Custom Column**
 - Name: TotalSpent
 - Formula: [Quantity] * [Price]
 - Click **OK**

- **Group the Table by CustID**
- Select the **CustID** column
- Go to **Transform > Group By**
- In the dialog box:
 - **Group by:** CustID
 - **New column name:** CustomerTotal
 - **Operation:** Sum
 - **Column:** TotalSpent

- Click **OK**
- **Apply Changes**
- Click **Close & Apply** to load the grouped data into Power BI

13. Fix inconsistent date formats (e.g., 01/10/2023 vs. 2023-01-10) in OrderDate.

- **Open Power Query Editor**
- Go to **Home > Transform Data**
- **Select the OrderDate Column**
- Click on the column named **OrderDate**
- **Change the Data Type to Date**
- Go to **Transform > Data Type > Date**
- Power Query will attempt to interpret all values as valid dates, regardless of format
- **Handle Errors (if any)**
- If some rows show errors (), it means Power Query couldn't parse them
- To fix:
 - Select the column
 - Go to **Transform > Replace Errors**
 - Replace with a default date (e.g., #date(2023, 1, 1)) or null
- **Optional: Use Locale for Specific Formats**

- If your dates are stored in a specific regional format (e.g., dd/mm/yyyy), do this:
 - Right-click the column > **Change Type > Using Locale**
 - Choose **Date** and select the correct **Locale** (e.g., English (United Kingdom) for dd/mm/yyyy)

- **Apply Changes**

- Click **Close & Apply** to load the cleaned dates into Power BI

14. Create a conditional column: Label orders as "High Value" if Price > 100.

Open Power Query Editor

- a. Go to **Home > Transform Data**

Add a Conditional Column

- b. In the ribbon, select **Add Column > Conditional Column**

Configure the Logic

- c. **New column name:** ValueLabel (or any name you prefer)
- d. **Column name:** Price
- e. **Operator:** is greater than
- f. **Value:** 100
- g. **Output:** "High Value"
- h. **Else:** "Standard" (or leave blank if you prefer)

Click OK

- i. This creates a new column with labels based on the condition

Apply Changes

- j. Click **Close & Apply** to load the updated data into Power BI

15. Optimize the query to reduce refresh time (e.g., remove unused columns early).

1. Remove Unused Columns Immediately After Import

- As soon as you load your data (e.g., from SQL, Excel, or CSV), remove columns you don't need:
 - Select only the columns you want
 - Go to **Home > Remove Columns > Remove Other Columns**

This minimizes the amount of data Power BI processes in every subsequent step.

2. Avoid Complex Steps Before Column Reduction

- Don't apply filters, merges, or transformations on columns you'll eventually discard.
- Every step adds processing time—so trim the fat first.

3. Use Column Selection in Source Queries (Advanced)

- If you're connecting to a SQL database, write a custom SQL query that selects only the needed columns:

```
SELECT CustID, OrderDate, Quantity, Price FROM Sales
```

4. Disable Auto Data Type Detection

- Power BI tries to guess column types, which can slow things down.
- Go to **File > Options > Data Load > Type Detection** and set it to "**Do not detect column types**" if you prefer manual control.

5. Avoid Unnecessary Steps

- Review the **Applied Steps** pane and remove any unused or redundant steps.
- Combine steps where possible (e.g., filtering and renaming in one go).